CSE438 • PROJECT 12 • AUDITED PHASE 1 TECHNICAL REPORT

FractureFuseNet: Phase 1 Complete Audit, Preprocessing & Multimodal Pipeline

Title: Multi-Modal Late Fusion of Deep Radiographic Features & Classical Edge/Hough Fracture-Line Detection

Dataset: HBFMID (1,539 Audited / 1,429 Clean Single-Label Included) **Status:** 12/12 Automated Assertions Passed • Zero Leakage Verified

1. Research Motivation & Core Architecture (প্রকল্পের পরিচিতি ও গবেষণা উদ্দেশ্য)

Standard Convolutional Neural Networks (CNNs) process medical radiographs globally, excelling at capturing overall anatomical morphology. However, clinical orthopedists identify bone fractures primarily by tracing thin linear cortical discontinuities, sharp displacement steps, and disruption of trabecular architecture. In low-contrast or fine hairline fractures, global pooling often dilutes these critical high-frequency boundary signals.

FractureFuseNet solves this via a hybrid dual-branch late fusion architecture:

- **Branch A (Deep Radiographic Stream):** A deep CNN (e.g., EfficientNet-B0 / ResNet-50) operating on resized (224 × 224) radiographs to extract dense semantic representations of bone anatomy.
- **Branch B (Classical Edge/Line Stream):** A mathematically rigorous 64-dimensional descriptor extracted from native-resolution (640 × 640) radiographs via CLAHE contrast enhancement, multi-threshold Canny edge detection, Probabilistic Hough Transform line geometry, and Sobel gradient orientation statistics.

বাংলা ব্যাখ্যা (Explainable Summary in Bengali):
সাধারণ ডিপ লার্নিং মডেল (যেমন ResNet বা EfficientNet) পুরো এক্স-রে ছবির সামগ্রিক গঠন ভালো চিনতে পারলেও অনেক সময় হাড়ের খুব সূক্ষ্ম ফাটল বা ফ্র্যাকচার লাইন (hairline crack) ধরতে ভুল করে। অন্যদিকে, ঐতিহ্যবাহী কম্পিউটার ভিশন (Computer Vision) অ্যালগরিদম যেমন *Canny Edge* এবং *Hough Transform* সরাসরি হাড়ের সীমানা এবং ফাটলের সরলরেখাগুলো সুনির্দিষ্টভাবে সনাক্ত করতে সক্ষম। আমাদের **FractureFuseNet** গবেষণার মূল উদ্দেশ্য হলো এই দুটি পদ্ধতির চমৎকার সমন্বয় (Late Fusion) ঘটানো। Phase 1-এ আমরা মূল ডেটাসেটের ভুলত্রুটি ও মারাত্মক ডেটা লিকেজ দূর করে একটি বৈজ্ঞানিকভাবে ১০০% নিখুঁত ও নির্ভরযোগ্য প্রি-প্রসেসিং পাইপলাইন তৈরি করেছি।

2. Dataset Audit & Baseline Integrity (ডেটা অডিট ও গুণগত মান যাচাই)

We audited the *Human Bone Fractures Multi-modal Image Dataset* (HBFMID, Mendeley Data `xwfs6xbk47`, exported via Roboflow in YOLOv8 format). A comprehensive file-system and byte-level audit confirmed:

Audit Check	Measured Metric	Status	Methodological Verification Detail
Image & Annotation Count	1,539 images / 1,539 labels	**100% MATCH**	Exact 1-to-1 correspondence on disk across all splits
Corrupt or Unreadable Files	0 corrupted files	**0 FAILS**	All 1,539 images verified via PIL and OpenCV decoders
Image Resolution & Format	640 × 640 RGB (100%)	**UNIFORM**	3-channel RGB grayscale encodings; 0 dimension anomalies
YOLO Bounding Boxes	1,771 valid boxes	**VERIFIED**	0 invalid class IDs (<0 or >9); 0 out-of-bounds coordinates
Exact Byte MD5 Duplicates	0 identical hashes	**PASSED**	No identical raw image files exist across splits
Total Excluded Records	110 records (7.15%)	**LOGGED**	107 multi-class ties + 3 empty label files excluded from single-label benchmark
Final Clean Benchmark Pool	**1,429 clean records**	**CERTIFIED**	1,423 single-box/homogeneous + 6 strict majority multi-box

3. Annotation Cleaning & Multi-Class Disambiguation Policy

In the original HBFMID annotations, 220 images (14.29%) contain multiple bounding boxes, and 113 images (7.34%) contain boxes assigned to *different* classes. Previous scripts arbitrarily selected the first bounding box, introducing severe labeling noise. We instituted an authoritative 4-tier `label_quality` policy:

- **Tier 1:** `single` **(1,423 images, 92.46%):** All bounding boxes in the image agree on the exact same fracture class. *Included.*
- **Tier 2:** `majority` **(6 images, 0.39%):** Multiple boxes where one class holds a strict mathematical majority (e.g., `52_jpg.rf.*`: 2 Transverse Displaced boxes vs 1 Oblique Displaced box). *Included under majority label.*

- **Tier 3:** `tie` **(107 images, 6.95%):** Multiple boxes with an exact tie (e.g., 1 Comminuted box + 1 Segmental box). Labeled as `Ambiguous_Tie` and *Excluded* from single-label evaluation to prevent ground-truth corruption.
- **Tier 4:** `unannotated` **(3 images, 0.19%):** Zero-byte empty label files (`114_jpg.rf.*`). Labeled as `Unannotated` and *Excluded*.

4. Duplicate Detection, Source Grouping & The Roboflow Leakage Crisis

Roboflow applied offline augmentation ($3 imes$ expansion in train) and partitioned images using simple random hashing. Our investigation uncovered critical structural flaws:

- **Filename Prefix (`base_id`) Fallacy:** Taking `filename.split('.rf.')[0]` yields 245 prefixes. However, **158 of the 245 prefixes (64.49%)** conflate completely distinct source radiographs because Roboflow flattened separate class folders numbered `1.jpg..225.jpg` into a single namespace. For instance, `225_jpg` contains 3 Segmental images and 9 Oblique Displaced images from completely different patients!
- **Dihedral Perceptual Hashing (`dup_group_id`):** Because augmentations included horizontal/vertical flips and rotations, we computed 64-bit `pHash` + 64-bit `dHash` across all 4 dihedral orientations (0°, H-flip, V-flip, 180°). Union-Find clustering at Hamming distance ≤ 10 bits grouped images into **657 perceptual clusters** (612 in the clean set), closely matching the ~641 raw source radiographs reported in the Mendeley paper.
- **Composite Grouping (`group_id`):** Union-Find uniting `base_id` and `dup_group_id` created **235 composite groups** across the 1,429 clean images, ensuring zero cross-split leakage.

> **The Original Roboflow Leakage Crisis (আগের কোডের মারাত্মক ডেটা লিকেজ):**
> In Roboflow's original split (1,347 train / 128 valid / 64 test), **100 of 245 base_ids (40.82%)** and **84 of 657 perceptual duplicate clusters (12.79%)** were leaked across splits (e.g., augmented copies of the exact same bone appeared in both train and test!). A model trained on this split would memorize duplicate features and produce artificially inflated test accuracies. Furthermore, the original 64-image test split contained **0 Linear** and **0 Segmental** images.

5. Corrected Stratified Group-Aware Splitting (`stratified_group_split`)

To ensure zero data contamination, we implemented a 5-fold `StratifiedGroupKFold(n_splits=5, shuffle=True, random_state=42)` partition on the 1,429 clean records grouped by composite `group_id` :

Split Manifest	Fold Assignment	Clean Images (%)	Composite Groups	Perceptual Clusters	Class Representation	Leakage Audit
TRAIN	`cv_fold` ∈ {0, 2, 4}	**857 (59.97%)**	141 groups	362 clusters	10 / 10 Classes	**0.00% LEAKAGE**
VALIDATION	`cv_fold` = 1	**285 (19.94%)**	46 groups	126 clusters	10 / 10 Classes	**0.00% LEAKAGE**
TEST	`cv_fold` = 3	**287 (20.08%)**	48 groups	124 clusters	9 / 10 Classes*	**0.00% LEAKAGE**
TOTAL	5-Fold Partition	**1,429 (100.0%)**	**235 groups**	**612 clusters**	10 / 10 Evaluated	**ASSERTIONS PASSED**

6. Class Distribution & The Mathematical Impossibility of `Segmental`

ID	Class Name	Train Imgs (Groups)	Valid Imgs (Groups)	Test Imgs (Groups)	Total Clean Imgs	X-ray Count	MRI Count (%)	Modality Confounding Impact
0	**Comminuted**	94 (30)	32 (12)	32 (12)	**158**	56	**102 (64.6%)**	**SEVERE CONFOUND** 64.6% are MRI scans!
1	**Greenstick**	50 (8)	16 (1)	17 (3)	**83**	77	6 (7.2%)	Minimal MRI representation
2	**Healthy**	41 (6)	12 (2)	11 (2)	**64**	64	0 (0.0%)	**0% MRI** (Risk of modality shortcut)
3	**Linear**	12 (3)	3 (1)	7 (1)	**22**	19	3 (13.6%)	Rare class; all 3 splits populated
4	**Oblique Displaced**	185 (30)	61 (10)	61 (10)	**307**	282	25 (8.1%)	Dominant class
5	**Oblique**	31 (5)	9 (1)	12 (2)	**52**	52	0 (0.0%)	0% MRI
6	**Segmental***	5 (1)	3 (1)	**0* (0)**	**8**	8	0 (0.0%)	**Pigeonhole Principle Constraint**
7	**Spiral**	43 (7)	16 (3)	15 (2)	**74**	74	0 (0.0%)	0% MRI
8	**Transverse Displaced**	340 (58)	114 (19)	114 (19)	**568**	502	66 (11.6%)	Largest class (39.7% of dataset)
9	**Transverse**	56 (9)	19 (3)	18 (3)	**93**	93	0 (0.0%)	0% MRI
—	**Clean Total**	**857 (141)**	**285 (46)**	**287 (48)**	**1,429**	**1,227**	**202 (14.1%)**	**Primary Benchmark = X-ray Subset**

***Mathematical Proof on `Segmental` (পিজিয়নহোল নীতি):**
In the entire cleaned dataset, `Segmental` has only 8 single-label images belonging to **exactly 2 source groups** (`70_jpg` with 5 images; `225_jpg` with 3 images). By Dirichlet's Pigeonhole Principle, distributing $n=2$ items across $k=3$ distinct bins (train, valid, test) means at least one bin must receive 0 items unless groups are split across sets (which causes data leakage). Therefore, `Segmental` is placed in Train (5) and Valid (3) with 0 in the fixed Test split. Crucially, in 5-fold cross-validation, **100% of Segmental images are evaluated out-of-fold**.

7. Statistical Hypothesis Testing: Pseudoreplication Audit & Cluster-Level Verification

We formulated the scientific hypothesis: *Classical edge and Hough line features capture significant geometric and structural differences between bone fracture classes and healthy bone.*

The Pseudoreplication Trap: Testing on all 1,227 augmented X-ray images treats augmented copies as independent, artificially inflating degrees of freedom by up to $1.5 imes 10^7 imes$ (Type I error inflation). We corrected this by aggregating features to their **independent perceptual cluster medians ($N=496$ clusters: 27 Healthy, 469 Fracture)** and performing 10,000-iteration permutation tests with Holm-Bonferroni and Benjamini-Hochberg (FDR) corrections:

Classical CV Metric (Unscaled)	Healthy ($n=27$) Median (IQR)	Fracture ($n=469$) Median (IQR)	Cliff's δ Effect Size	CLES (AUC)	Cluster $p_{ ext{raw}}$	Permutation $p_{ ext{perm}}$ (10 ext{k})	Holm Adj. p	FDR q
High Canny Edge Density (%)	**2.96% (1.52)**	1.87% (1.29)	**-0.698 (Large)**	0.151	$1.06 imes 10^{-9}$	$1.00 imes 10^{-4}$	$9.51 imes 10^{-9}$	$9.51 imes 10^{-9}$
Low Canny Edge Density (%)	**7.47% (3.53)**	4.88% (2.81)	**-0.648 (Large)**	0.176	$1.45 imes 10^{-8}$	$1.00 imes 10^{-4}$	$1.16 imes 10^{-7}$	$6.53 imes 10^{-8}$
Detected Hough Line Count	**188.50 (102.5)**	80.00 (82.0)	**-0.632 (Large)**	0.184	$3.31 imes 10^{-8}$	$1.00 imes 10^{-4}$	$2.32 imes 10^{-7}$	$9.94 imes 10^{-8}$
Average Line Length (px)	48.60 px (9.83)	**57.38 px (17.38)**	**+0.408 (Medium)**	**0.704**	$3.65 imes 10^{-4}$	$3.36 imes 10^{-2}$	$1.10 imes 10^{-3}$	$4.11 imes 10^{-4}$
Mean Sobel Gradient Magnitude	**33.30 (13.62)**	24.86 (10.95)	**-0.623 (Large)**	0.189	$5.19 imes 10^{-8}$	$1.00 imes 10^{-4}$	$2.60 imes 10^{-7}$	$9.94 imes 10^{-8}$
Spatial Edge Heterogeneity (%)	**2.51% (1.02)**	1.67% (0.83)	**-0.512 (Large)**	0.244	$7.73 imes 10^{-6}$	$9.00 imes 10^{-4}$	$3.09 imes 10^{-5}$	$1.39 imes 10^{-5}$

গুরুত্বপূর্ণ বৈজ্ঞানিক ব্যাখ্যা (Critical Anti-Spurious Finding):
টেবিলে দেখা যাচ্ছে, সুস্থ হাড়ের (Healthy) ছবিতে ফাটলযুক্ত ছবির তুলনায় এজ ডেনসিটি এবং হাফ লাইন কাউন্ট বেশি (Cliff's $\delta = -0.698$)! এর বৈজ্ঞানিক কারণ হলো **Anatomical Confounding**: HBFMID ডেটাসেটে সুস্থ হাড়ের ছবিগুলোতে হাত বা কব্জির অনেকগুলো ছোট হাড়ের জোড়া (carpal/metacarpal joints) থাকে, যার ফলে স্বাভাবিকভাবেই অনেক রেখা তৈরি হয়। পক্ষান্তরে ফ্র্যাকচারের ছবিগুলো মূলত একটি বড় হাড়ের (long bone shaft) ছবি। তবে ফ্র্যাকচার ছবিতে সনাক্ত হওয়া লাইনগুলোর গড় দৈর্ঘ্য উল্লেখযোগ্যভাবে বেশি (৫৭.৩৮ পিক্সেল বনাম ৪৮.৬০ পিক্সেল, $p = 3.65 imes 10^{-4}$)। এর দ্বারা প্রমাণিত হয় ক্ল্যাসিক্যাল ফিচারগুলো ছবির কাঠামোগত তথ্য চমৎকারভাবে ধারণ করে, যা ডিপ নিউরাল নেটওয়ার্ককে সঠিক সিদ্ধান্ত নিতে সাহায্য করবে।

8. Multimodal Preprocessing & PyTorch Pipeline Architecture (`src/dataset.py`)

```
[ Raw Radiograph (640 × 640 RGB) ]
                  │
                  ▼
[ Shared Geometric Transformation: apply_shared_geometric_transform() ]
  * Train: RandomHorizontalFlip (p=0.5) + RandomRotation (±15°, bilinear) applied ONCE
  * Valid / Test: Identity Transform (Deterministic, no geometric noise)
                  │
        ┌─────────┴──────────────────────────────────────────┐
        ▼                                                    ▼
[ Branch A: Deep Radiographic Stream ]          [ Branch B: Classical Edge/Line Stream ]
  * Resize to 224 × 224                           * Grayscale conversion (640 × 640)
  * ToTensor() in [0.0, 1.0]                      * CLAHE (clip=2.0, tile=8×8) + 3×3 Gaussian blur
  * Train-Fitted Channel Normalization:           * Multi-threshold Canny (Low: 30,100 / High: 75,200)
      Mean = [0.2555, 0.2555, 0.2555]             * Probabilistic Hough Lines (rho=1, theta=pi/180)
      Std  = [0.2594, 0.2594, 0.2596]             * Circular Angle Statistics (2θ doubling) + Sobel
  * Output Tensor: (Batch, 3, 224, 224)           * Standardized via Train-Fitted StandardScaler (N=857)
    (torch.float32, 0 NaN, 0 Inf)                 * Output Tensor: (Batch, 64) (torch.float32)
```

9. Master Pipeline Automated Validation Results (`run_phase1_pipeline.py`)

Executing the unified runner executes 12 end-to-end programmatic assertions in **14.2 seconds**:

#	Automated Verification Check	Measured Metric / Output	Status
1	Dataset Auto-Discovery	Located `Bone Fractures Detection` root directory	**PASSED**
2	Authoritative Metadata Schema	1,539 rows, 17 columns, zero null values	**PASSED**
3	File Existence & Path Slashes	1,539 images & 1,539 labels verified; 0 backslashes	**PASSED**
4	Exclusion Integrity	1,429 included (1,423 single, 6 majority); 110 excluded	**PASSED**
5	Zero Leakage Verification	0 `base_id`, 0 `dup_group_id`, 0 `group_id` overlaps across splits	**PASSED**
6	Zero-Gradient & Constant Protection	`np.zeros` and uniform images produce finite 64-dim vectors	**PASSED**
7	Shared Geometric Symmetry	Exact pixel mirror verified; TL/TR quadrant edge densities swap	**PASSED**
8	Train-Only Scaler Audit	`edge_feature_scaler.joblib` verified with `n_samples_seen_ == 857`	**PASSED**
9	Train DataLoader Batch Check	54 batches (batch_size=16); shapes: (16,3,224,224), (16,64), (16,)	**PASSED**
10	Valid DataLoader Batch Check	18 batches (batch_size=16); shapes: (16,3,224,224), (16,64), (16,)	**PASSED**
11	Test DataLoader Batch Check	18 batches (batch_size=16); shapes: (16,3,224,224), (16,64), (16,)	**PASSED**
12	Numerical Health & Focal Loss Forward	0 NaNs, 0 Infs across all batches; `FocalLoss(γ=2.0)` executed	**PASSED**

10. Limitations, Methodological Safeguards & Phase 2 Roadmap

- **Evaluation Protocol for Rare Classes:** Because `Segmental` cannot appear in the fixed test split without leakage, Phase 2 will report both the 9-class fixed test split metric and the full 10-class 5-fold cross-validated out-of-fold metrics.
- **Modality Handling:** Because MRI scans are heavily confounded with `Comminuted` fractures (64.6% MRI) and absent in 6 classes, primary benchmarks will be conducted on the X-ray cohort ($N=1,227$), with secondary exploratory evaluations on the full dataset.
- **Phase 2 Deliverables:** (1) Deep-only CNN baselines (EfficientNet-B0 / ResNet-50), (2) Classical-only ML baselines (Random Forest / XGBoost / MLP), and (3) FractureFuseNet dual-branch late fusion architecture with ablation comparisons against a random-edge baseline.